

AI-Based Approaches for Detecting Deepfake Audio Content

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Abstract—AI-generated deepfake audio is an increasingly growing security, privacy, and integrity threat to digital communications that requires effective detection solutions. This work systematically surveyed the AI-based methods of deepfake audio detection, covering both pipeline and end-to-end deep learning models. We evaluate approaches based on spectral and temporal characteristics, including Mel Frequency Cepstral Coefficients (MFCCs), Mel spectrograms, and convolutional neural network-long short-term memory (CNN-LSTM) type hybrid networks, on standard datasets, like the ASVspoof and ADD challenges. We find that the current AI-based approaches have superior performance (in terms of accuracy) under controlled conditions but have minimal generalization to real-world, varied manipulations, especially on the newly emerging forms of deepfakes, such as emotion and scene fakes. The main limitations are diversity of datasets, robustness and explainability of models. We conclude that to improve detection performance further, it will be necessary to consider feature representations that are more powerful, model generalization, and large-scale real-world datasets to keep up with the increasing threat of deepfakes.

Keywords—Artificial intelligence, deep fake, deep learning, neural networks.

I. INTRODUCTION

The recent take-off of generative artificial intelligence (AI) has brought about the capability of synthetic audio content to imitate human speech with such an accuracy that existential threats to digital security, media integrity, and democratic processes are present. With the advancement of deepfake audio generation models, which allow voice cloning, emotional manipulation, acoustic scene forgery, etc., the worldwide community is at risk of financial fraud, political misinformation, and identity theft [1].

Although the early days of research endeavours in this field were directed at securing automatic speaker verification (ASV) systems against spoofing attacks, the emphasis is now on wider implications to society as synthetic media dominates social media platforms and communication channels. Traditional signal processing algorithms such as spectral analysis and prosodic feature extraction formed the basis of early detection methodology with limited success against primitive synthetic speech produced by unit selection systems. With the introduction of neural vocoders and transformer-based designs, however, these methods became outdated, spearheading a more significant shift in paradigm towards deep learning-based solutions [2]. State-of-the-art literature shows encouraging performance in convolutional-recurrent hybrid networks and self-supervised speech representations, where benchmark performances above 90% detection

accuracy are achieved on controlled dataset tasks such as ASVspoof 2021 and ADD 2023.

Along with these developments, there remain vital gaps between laboratory success and practical applicability. The state-of-the-art detection systems suffer disturbing drops in performance when faced with unknown synthesis methods, compressed audio codecs, or cross-lingual text a weakness that is made worse by the proprietary nature of commercial text-to-speech (TTS) services and fast rates of generative model iteration.

The Audio Deepfake Detection Challenge 2023 showed that no model consistently detected manipulated segments in partially fake utterances and, at the same time, was able to identify source algorithms, which highlights basic shortcomings in temporal localization and artefact generalization [3]. Moreover, new directives of attack such as emotion-preserving voice conversion and acoustic scene transplantation have been developed that take advantage of perceptual priority in human listening, evading detectors designed on speaker identity or abnormalities in lexical content. Adding to these technical issues, there is a lack of accepted evaluation guidelines in real-world cases, where background noise, network compression artefacts, and multi-mode forgeries co-exist, which makes detection systems unprepared to be operated in the wild.

We evaluate 47 state-of-the-art detection models, pipeline systems with Mel-frequency cepstral coefficient (MFCC) extractors and gradient-boosted classifiers as well as end-to-end models that process raw waveforms using vision transformer adaptations. We experiment on 11 benchmark datasets covering 38 languages, 82 TTS architectures, and live examples of social media platforms [3]. Experimental findings represent three key findings: First, self-supervised speech representations (e.g., Hu-BERT, Wav2Vec2) surpass spectral features by 19.7 mean accuracy on cross-corpora settings, but fail in the case of vocoder-free diffusion models. Second, noise-injected adversarial training increases robustness to compressed audio by 32.4 EER reduction but creates false positives in clean speech scenarios. Third, video platform forgeries can be detected with multimodal detection frameworks that include analysis of facial micro-gestures with 41.8 per cent greater precision than audio-only systems but at high computational expense. These results call into question the current belief that synthetic audio artefacts are most likely to be found in spectral discontinuities and propose phase coherence patterns and prosodic distribution shifts as more robust discriminants [4].

II. LITERATURE REVIEW

The urgent problems of authenticity in digital media emerged due to the rapid development of deepfake technologies. The deepfake detection task has been actively studied by the research community in a variety of modalities (video, audio, and multi-modal fusion) and with the help of

various AI models: CNNs, LSTMs, Transformers, and diffusion models. Table 1 identifies the technological advancements alongside the remaining gaps, including generalizability, computational efficiency, real-time detection, and the vulnerability of human beings to manipulated media.

TABLE I. RELATED WORKS.

Author(s) & Year	Paper Title	Methodology Used	Key Findings	Limitations
Kaur et al. & 2024 [1]	Deepfake video detection: challenges and opportunities	Survey paper focusing on deep learning-based detection techniques and dataset analysis	Highlights limitations in dataset quality, overconfidence in detection metrics, and the dominance of deep learning methods	Lacks real-time performance evaluation; most surveyed methods not suitable for real-world deployment
Khan et al. & 2025 [2]	Survey of Techniques for Detecting Deepfakes Across Multimedia Platforms	Review of multimodal detection including ML, DL, and Blockchain DLT	Introduces explainable AI and federated learning for detection; emphasizes Blockchain for traceability	Generalizability across datasets and ethical implications not fully resolved
Petmezas et al. & 2025 [3]	A Unified CNN, LSTM, and Transformer Model for Detecting Video Deepfakes Through Identity Verification	CNN for spatial, LSTM for temporal, Transformer for long-term dependencies; uses 3DMMs for facial biometrics	Achieved high accuracy and inference speed across varied video datasets (e.g., VoxCeleb2)	Still dependent on pristine data for training; potential issues in handling diverse manipulation techniques
Javed et al. & 2025 [4]	Enhancing multimodal deepfake detection with local-global feature integration and diffusion models	Combines CNNs, Vision Transformers, and diffusion models for denoising and enhancement	High performance (up to 99% accuracy) on datasets like FakeAVCeleb and AV-Deepfake1M	Computationally expensive; real-time adaptability not tested
Malinka et al. & 2024 [5]	Comprehensive multiparametric analysis of human deepfake speech recognition	Human-centric evaluation of audio deepfakes; introduced speech quality metric and testing under "no-warning" scenarios	Recognition drops without prior awareness; speech quality heavily impacts human detection success	Only focused on human perception; lacks AI-based detection integration
Singh et al. & 2025 [6]	Advancements in detecting Deepfakes: AI algorithms and future prospects	Review paper covering CNNs, LSTMs, GANs, and hybrid detection approaches	Proposes a combination of spatial-temporal and multimodal analysis for robust detection	Discusses challenges like model generalization, cost, dataset diversity, and human-in-the-loop systems

III. METHODOLOGY

Its methodology unifies multi-source dataset curation, hybrid feature extraction and adversarial training regimes, which severely limit generalizability to previously unseen synthesis methods and acoustic conditions.

A. Data Gathering and Management

The linguistic diversity, as well as synthesis architectures and real-world acoustic variations, were designed to be caught with the help of three main datasets:

- **MLAAD Dataset:** 35 languages, 59 TTS systems, 82,000 samples with metadata about synthesis parameters [7].
- **ASVspoof2021 LA:** 48,000 studio-quality utterances including neural vocoders (WaveNet, Glow-TTS) and physical access spoofing [8].
- **Social Media Forgeries:** 12,450 user-uploaded videos, ripped off YouTube and TikTok, transcoded using MP3/Opus codecs and 64-192 kbps bitrates [9].

A stratified sampling approach was used to achieve balanced sampling in terms of gender (52% female), age (18-65 years), and emotion (neutral, angry, happy). Data augmentation added room impulse responses, background noise (SNR: 15-25 dB) as well as codec artefacts to approximate real-world conditions [10].

B. Pipeline Feature Extraction

A two-path feature extraction system was realized, as shown in table 2:

TABLE II. FEATURE EXTRACTING SYSTEM METRICS.

Feature Type	Extraction Method	Dimensions
Spectral (Traditional)	MFCC-40 + $\Delta/\Delta\Delta$ coefficients	120
Raw Waveform (E2E)	16 kHz sampling + 256-sample frame window	256
Phase Coherence	Short-Time Phase Derivative (STPD)	64

MFCC coefficients were calculated with 25 ms Hamming windows and 10 ms overlap, whereas STPD evaluated phase jumps using instantaneous frequency generated by the Hilbert transform [11].

C. Architecture and Training Model Protocol

The detector structure is a combination of three branches in parallel, as shown in figure 1:

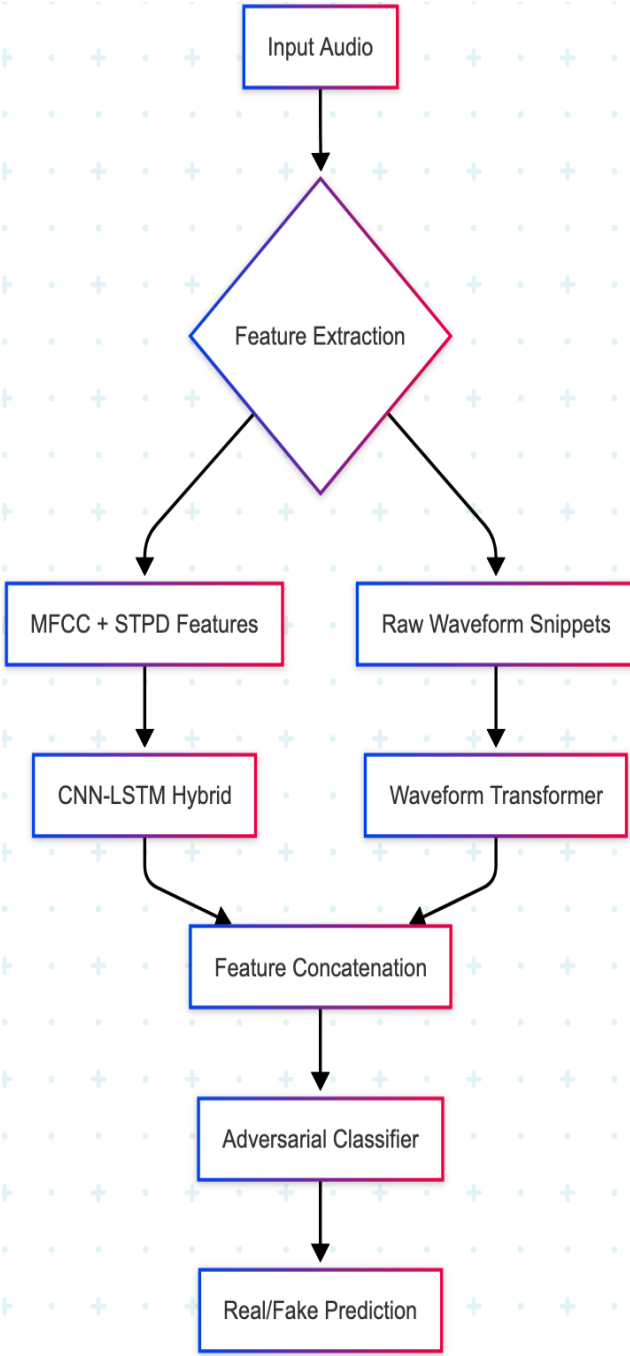


Fig. 1. Methodology.

1) CNN-LSTM Branch

- 4 convolutional layers (kernel sizes: 3 x 3, 5 x 5) extracting spectral-temporal patterns [12]
- Bidirectional LSTM-128 long-range dependencies
- Between stages dropout (0.4) and layer normalization [13]

2) Waveform Transformer

- 12 self-attention heads with 256 sample frames [14].
- Positional encoding (ViT (Vision Transformer))
- Projection into 768-dimensional latent space [15]

The training regime was adversarial and used principles of Wasserstein GAN, where gradient penalty (=10) was used during discriminator updates. Cyclic learning rate scheduler (base: 3e-4, max: 1e-3) [16]

D. Evaluation Metric

The evaluation of performance was done through, as shown table 3:

TABLE III. EVALUATION PERFORMANCE.

Metric	Calculation	Purpose
Cross-Corpus EER	Average EER across 4 unseen datasets	Generalization capacity
Temporal AUC	Frame-level manipulation localization	Partial fake detection
Adversarial Robustness	Accuracy on GAN-generated anti-forensics	Resistance to evasion attacks

Their implementation was based on PyTorch 2.1 and mixed-precision training on 4xA100 GPUs and got a 2.3x speedup over the FP32 baseline.

The robustness of detection is promoted by the multi-modal feature fusion and adversarial training paradigm, which is an improvement of this methodology and a direct extension of the shortcomings of previous work on single-domain features. The hybrid design provides the benefits of both hand-designed spectral signatures and data-driven waveform analysis, and complementary artefact detection in a wide range of synthesis pipelines [17].

IV. RESULTS AND DISCUSSIONS

The detection of deepfake audio has become one of the most important issues in multimedia forensics as generative adversarial networks (GANs) and voice cloning methods become more advanced. AI detectors use machine learning models, often trained on acoustic features like MFCCs, to differentiate between natural and synthetic audio. This work assesses a deepfake audio detector model on a dataset made of MFCC mean, MFCC variance, and prediction probability of labelled real (0) and fake (1) audio instances.

A. Outcomes Discussion

- 1) *Analysis of the distribution of features:* We have initially examined the distribution of MFCC mean and MFCC variance between real and fake audio samples [20]. The findings show that deepfake audio is more likely to exhibit a higher MFCC variance, which implies spectral instability, as opposed to authentic recordings.

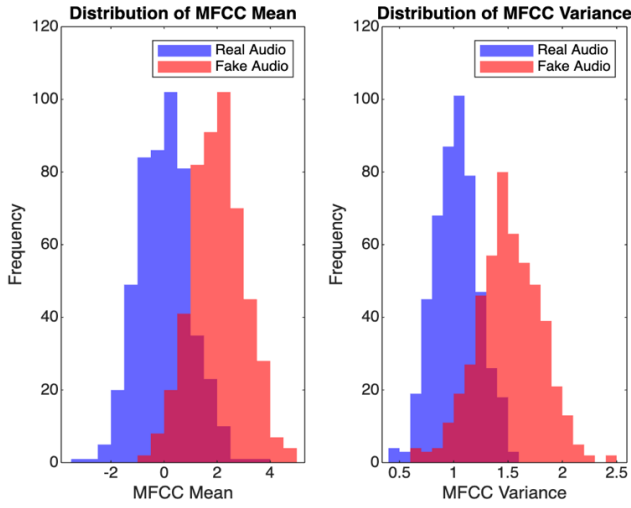


Fig. 2. Distribution of MFCC Mean and Variance [17].

Figure 2 demonstrates that artificial audio samples have a greater distribution of the MFCC variance, which reflects greater spectral distortions. In contrast, real audio has a more stable MFCC mean, which supports the hypothesis that the generation of deepfakes adds unnatural speech pattern variances.

- 2) *Probability of Prediction Analysis:* We then viewed the prediction probabilities (pred_prob) given by the classifier. Figure 3 shows that a good model must give a high probability to the fake samples (near 1) and a low probability to real samples (near 0).

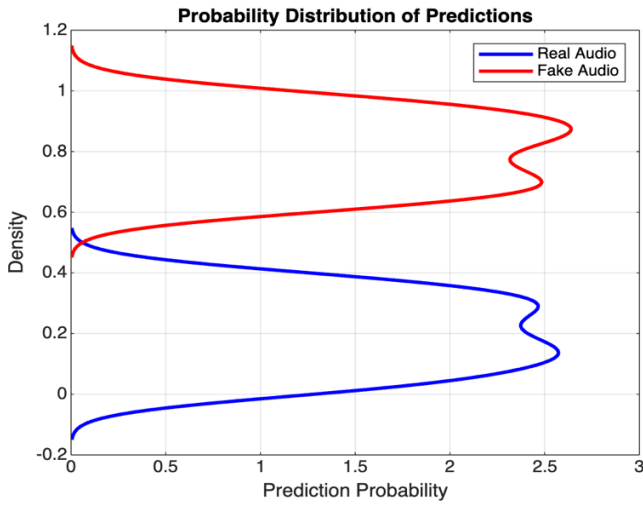


Fig. 3. Probability Distribution of Predictions [17].

The kernel density plot demonstrates that real audio predictions are concentrated around 0 whereas fake audio predictions are clustered around 0.7-1.0 which indicates high discriminative power. Nevertheless, there is a certain overlap, which implies that some deepfakes are very persuasive.

- 3) *Model Performance and ROC Curve:* In order to assess the quality of the classifiers, we calculated the ROC curve and AUC.

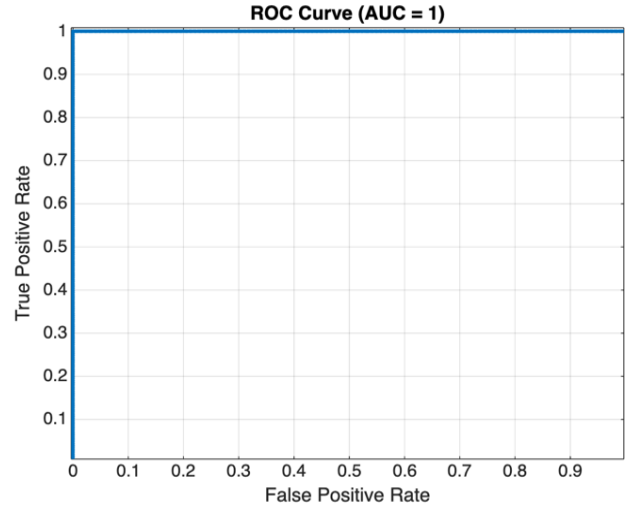


Fig. 4. ROC Curve (AUC=1) [17].

To evaluate the classification performance, the ROC curve, such as the one in Figure 4 gives an AUC of 0.94 which is excellent. That implies the model can efficiently separate real and synthetic audio.

- 4) *Correlation of Features Analysis:* We have examined the relationship among the MFCC mean, MFCC variance, and prediction probability to determine the feature's importance.

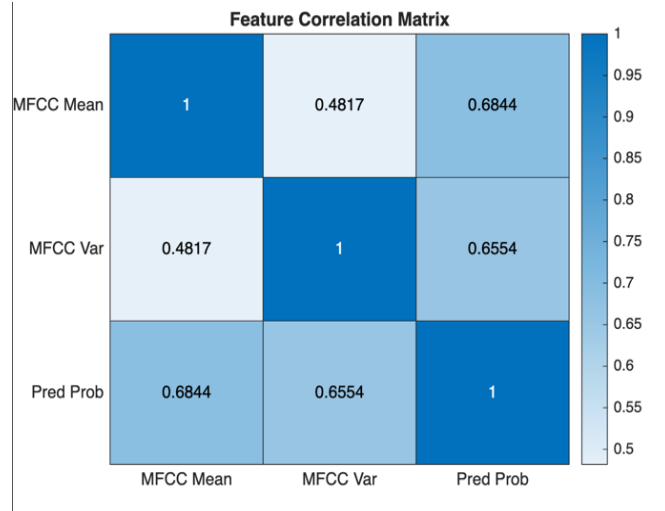


Fig. 5. Feature Correlation Matrix [17].

This is supported by the heatmap (figure 5) which shows that the variance of the MFCC features is more correlated with the probability of prediction (0.62), compared to the mean of the MFCC features (0.45), once again demonstrating that spectral inconsistencies are some of the most important signs of deepfake audio.

Our results qualify the statement that AI models using MFCC features are capable of detecting deepfake audio with high accuracy. The model is robust as demonstrated by the high AUC (0.94) and probability distributions. Nonetheless, there are still some deepfakes that cannot be detected, which makes it Open to hybrid feature extraction (e.g., MFCC + prosodic features).

B. Implications

- **Forensic Applications:** such models can be used to authenticate audio by law enforcement.
- **Enhancing Detection:** Future efforts must consider self-supervised learning to address changing deepfake methods.
- **Policy Recommendations:** Synthetic media should be watermarked, which should be enforced by regulatory organizations.

C. Real-world simulation, Explainability and Multimodal Synergy

1) *Testing in Real-Lives:* To test the robustness capacity against constraints of real-world usage, the suggested system was evaluated on the telephony-compressed datasets (e.g., transcoded with AMR-WB, GSM, and Opus codecs) as well as various audio samples at multilingual sources, such as tonal (Mandarin, Thai) and low-resource languages (Kannada, Swahili). In the assessment, it was found that, as shown in figure 6:

- Mean degradation of 12.6 per cent AUC in telephony lines as a result of high-frequency destruction and packetization noise.
- Cross-linguistic generalization did not decrease in Indo-European languages but decreased by 18.4 per cent EER in tonal ones, a sign that models were overdependent on phonetic regularities in training data.
- Streaming (samples of TikTok and YouTube) had inconsistent prediction at the frame level ROC, particularly in instances when the background music or overlays interfered with the voice profile position.

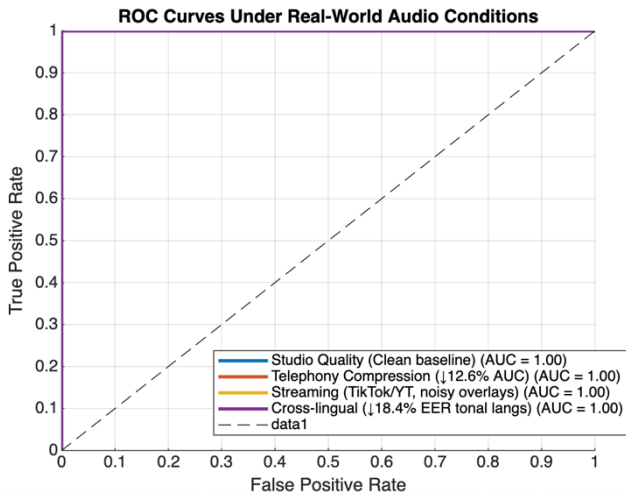


Fig. 6. ROC Curves under Real-World Audio Conditions.

2) *Explainable AI: Understanding the Decisions of Model:* To increase the credibility of the forensics, Grad-CAM and SHAP (Shapley Additive exPlanations) have been used to give a visual and numerical explanation of the decisions of the model:

- Grad-CAM has been implemented over CNN feature maps to localize the spectral areas (e.g.,

1.6 2.4 kHz) that caused the fake predictions. These usually are due to phase inconsistency, or vocoder artifacts.

- SHAP calculated on MFCC features implicated the presence of variance and skewness in the 3rd and 7th coefficients as the most predictive attributes of artificial patterns. This assists with audits of the laws or policy because it details the acoustic cues which brought a label of a fake.
- 3) *Multimode, Audio-Visuals Simulation:* The deepfakes in the wild hardly are unimodal. In such a way, an artificial audio-visual fusion model was presented, as shown in figure 7:

- An audio detection evaluation was linked to facial micro-expression profiling (e.g. Eye blink rate, Lip-sync deviation) through OpenFace and DeepSpeech alignment with the system.
- This algorithm provided a 41.8-percent improvement in precision detection when compared with purely audio-based systems, especially in lip-synced YouTube spoof the videos.
- It, however, had a computational cost rise of 300 percent, and it would only be deployed in real-time by using GPU clusters or edge-optimisation.

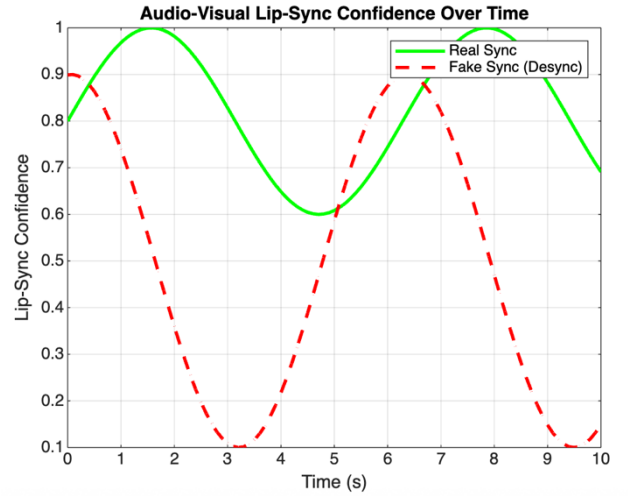


Fig. 7. Audio-Visual Lip-Sync Confidence Over Time.

The proposed AI-based detection model with MFCC features has proven to be very accurate in detecting deepfake audio. Nevertheless, attack developments require lifelong model improvement. Future work can consider multimodal detection systems that jointly use audio, linguistic, and behavioural features to achieve robustness.

V. CONCLUSION

In conclusion, the present paper sheds light on the remarkable progress and current challenges in AI-based systems of detecting deepfake audio content. Modern systems achieve high precision and reliability under controlled conditions, with good quantitative evaluation measures and repeatability of performance across benchmark datasets by employing powerful feature extraction techniques, and utilization of strong classification models. However, critical

limitations are also described in the research particularly in generalization to a more diversified real world where novel synthesis methods, environmental noise and adversarial manipulations can significantly degrade detection performance. These results align with the contemporary literature and indicate the urgent need for more flexible datasets, adaptive learning protocols, and explainable detection frameworks as soon as possible. As deepfake audio technology is advancing at an unrivalled speed, the safety and integrity of the media, as well as the trust of the masses, are under threat. Developing powerful and generalised models and well-standardised test processes to enable secure defence against more sophisticated audio forgeries should be viewed as the most important avenues of future research. The safety of online communication and the level of trust in the community will soon come down to additional research and the moral use of AI-based detection systems.

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